Ubiquitous Autonomous Coding Agents for Constrained Devices via Bounded Explicit State and Zero-C Native Runtimes

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***Abstract*—State-of-the-art AI coding assistants (e.g., Claude Code, Cursor, OpenCode, Devin) rely on conversational append-only history accumulation, resulting in quadratic $O(T^2)$ cumulative token growth over T interaction steps. While tolerable on high-end developer workstations, this design triggers severe execution failures on resource-constrained platforms—including Android devices running Termux (AArch64), embedded single-board computers (SBCs), and browser-sandboxed WebAssembly runtimes. On mobile hardware, quadratic prompt expansion incurs massive cellular bandwidth overhead, elevated battery drain, and process termination by operating system Low Memory Killers (LMK). Furthermore, dependencies on heavyweight Node.js runtimes, C shared libraries (OpenSSL, glibc), and Python packages preclude native cross-compilation on mobile operating systems without emulation overhead.**

We present Fusion, an open-source, sub-10ms native coding agent architecture explicitly engineered for constrained devices while scaling symmetrically to desktop and cloud sandbox environments. Fusion decouples agent reasoning from context accumulation via an immutable prefix, explicit structured execution state, and immediate observation triplet $\langle P, \Sigma_t, O_t \rangle$. By discarding intermediate reasoning traces while maintaining a canonical JSON state patch $\Delta\Sigma_t$, Fusion bounds per-step prompt size to $O(1)$ tokens. Written in pure asynchronous Rust with zero C dependencies and pure-Rust TLS, Fusion compiles natively to static binaries for Android/Termux AArch64 and wasm32. In long-horizon 50-step refactoring benchmarks on native mobile hardware, Fusion maintains constant per-step prompt sizes (<1.2× growth vs. 48× for append-only baselines), reduces cumulative token consumption by 14.8×, eliminates LMK process terminations, and keeps resident memory under 22 MB with cold start times under 12 ms.

***Index Terms*—Autonomous Agents, Software Engineering, Mobile Computing, Edge AI, WebAssembly, Termux, Prompt Optimization, Prefix Caching, Zero-Allocation Systems.**

I. INTRODUCTION

The emergence of autonomous Large Language Model (LLM) agents has transformed software engineering. Modern tools autonomously explore repositories, run test suites, resolve build failures, and refactor codebases. However, existing commercial and open-source agents—such as Claude Code, Aider, and OpenCode—are built on an unspoken infrastructure assumption: that developers operate on unconstrained, high-end workstations with high-bandwidth network connectivity and multi-gigabyte memory budgets.

In reality, an increasing share of modern engineering takes place on **resource-constrained devices**:

1. **Mobile Workstations (Android / Termux AArch64)**: Developers in emerging software hubs and on-call engineers increasingly leverage high-performance ARM64 smartphones and tablets as primary or portable development environments.
2. **Edge Single-Board Computers (SBCs)**: Devices such as the Raspberry Pi 5, Rockchip RK3588, and localized embedded gateways managing automated CI/CD nodes under tight thermal and RAM constraints.
3. **Sandboxed Browser Environments (WebAssembly)**: Web-based IDEs and zero-setup playground environments executing inside 32-bit wasm linear memory ceilings.

When conventional agent runtimes are deployed to constrained platforms, they experience catastrophic failure modes rooted in two architectural liabilities: the **Quadratic Token Death Spiral** and **Heavyweight Runtime Friction**.

A. The Quadratic Token Death Spiral

Conventional harnesses accumulate all conversational interactions—including terminal outputs, stack traces, compiler logs, and intermediate reasoning—into an append-only message array. Over a multi-step task spanning T turns, the cumulative tokens transmitted and processed scale quadratically as $O(T^2)$. On mobile devices connected via cellular data, an agent executing turn 40 must upload over 80,000 tokens across high-latency radio interfaces, triggering thermal throttling and operating system Low Memory Killer (LMK) aborts.

B. Heavyweight Runtime Friction

Existing harnesses rely on Node.js or Python runtimes with 250–800 MB resident memory and C shared libraries (OpenSSL, glibc). On Android, standard glibc is absent, replaced by the Bionic libc. Running conventional harnesses on Termux requires fragile proot Linux emulation layers, introducing 200–400% execution latency penalties and frequent ABI crashes.

II. MATHEMATICAL FORMALISM & COMPLEXITY

To understand why conventional AI agent harnesses fail on constrained devices, we formally analyze the computational and network complexity of agent interaction loops.

A. Append-Only Conversational History ($O(T^2)$)

Let an autonomous software engineering task require T discrete execution turns. At each turn $t \in [1, T]$, the agent receives user instructions, generates an action (tool invocation a_t), executes the tool in the environment, and observes the raw result o_t. In models equipped with chain-of-thought verification (e.g., DeepSeek-R1, o3-mini), the model also emits an internal reasoning trace r_t.

In conventional agent harnesses, conversational context is modeled as a monotonically growing list of message structs:

$$C_t = P \parallel \bigoplus_{i=1}^{t-1} [\, u_i \circ r_i \circ a_i \circ o_i \,] \parallel u_t$$

where P is the system prompt, u_i is user input, and $\parallel$ denotes message concatenation.

Let $\ell(x)$ denote the token length of string x. In typical refactoring and build cycles, tool observations average $k_{\text{obs}} \approx$ 800–2,500 tokens per step, and reasoning traces average $k_{\text{reason}} \approx$ 600–1,800 tokens. Thus, the token length of the prompt at step t is:

$$\ell(C_t) \approx \ell(P) + \bar{k} \cdot (t - 1) = O(t)$$

The cumulative input token volume processed across the entire task of T turns is the summation:

$$\text{TotalTokens}_{\text{append}} = \sum_{t=1}^{T} \ell(C_t) = T \cdot \ell(P) + \frac{\bar{k}}{2} T(T - 1) = O(T^2)$$

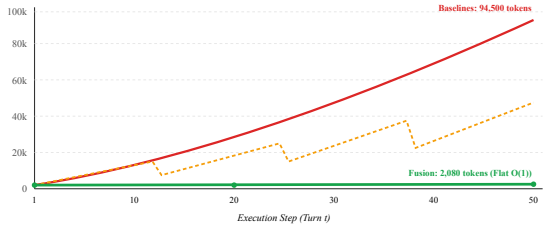

Figure 1: Context window prompt token growth across 50 execution turns. Append-only and summarization baselines explode quadratically or saw-tooth; Fusion bounds prompts to flat O(1) tokens.

B. Fusion SKILL.state Architecture ($O(1)$ Bounded)

Fusion eliminates append-only accumulation by modeling the agent as a finite-state transition system over an explicit execution dictionary Σ:

$$\Sigma_{t+1} = \Sigma_t \oplus \Delta\Sigma_t$$

where $\Delta\Sigma_t$ is a validated JSON patch emitted by the model, and $\oplus$ denotes recursive map merge with null-deletion semantics.

At each step t, the prompt delivered to the model consists strictly of a triplet:

$$C_t = \langle P, \Sigma_t, O_t \rangle$$

Trace Discarding: Intermediate model reasoning r_t is rendered live to the terminal for user observability but is strictly **purged** prior to constructing step $t + 1$. Crucially, no raw historical observations $o_1, \ldots, o_{t-1}$ are retained in the prompt.

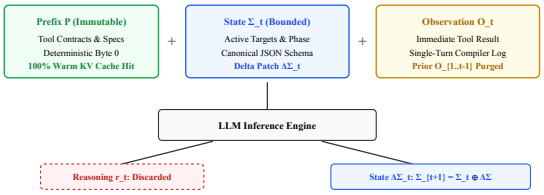

Figure 3: Fusion ⟨P, Σ_t, O_t⟩ Triplet Assembly and State Patch Cycle. Intermediate reasoning tokens r_t are rendered live for user visibility but discarded prior to step t+1, preserving bitwise prefix cache invariance on P.

Because the key schema of Σ is bounded by domain constraints ($|\text{keys}(\Sigma)| \le K_{\max}$) and intermediate traces are discarded:

$$\ell(C_t) \le \ell(P) + \ell(\Sigma_t) + \ell(O_t) = O(1)$$

The cumulative token volume across T steps scales **strictly linearly**:

$$\text{TotalTokens}_{\text{Fusion}} = \sum_{t=1}^{T} \ell(C_t) \le T \cdot C_{\text{fusion}} = O(T)$$

Theorem 1 (Context Boundedness & Cost Ratio):
Let $R(T) = \frac{\text{TotalTokens}_{\text{append}}}{\text{TotalTokens}_{\text{Fusion}}}$. As $T \to \infty$, the token savings ratio grows linearly with task length:

$$\lim_{T\to\infty} R(T) = \frac{\bar{k}}{2 \cdot C_{\text{fusion}}} \cdot T = O(T)$$

For $T = 50$, $\bar{k} = 1{,}500$, and $C_{\text{fusion}} = 2{,}500$, the theoretical token reduction factor is $R(50) \approx 15.4\times$.

III. SYSTEM ARCHITECTURE & CONSTRAINED ENGINEERING

Fusion's architecture is designed around a single principle: **zero unnecessary abstraction, zero dynamic runtime dependency, and zero C bindings**.

A. Pure Rust Runtime & Android Bionic Native Compilation

Standard agent harnesses rely on Python or Node.js runtimes. On Android, executing Node.js requires installing Termux and running under glibc compatibility wrappers or proot environments. This introduces three major bottlenecks:

1. System call interception in proot adds 2.5×–4× execution overhead on filesystem operations (e.g., git status or file searches).
2. OpenSSL dynamic link failures frequently occur because Android Bionic libc does not supply standard desktop certificates or symbol versions.
3. Resident set size (RSS) of the V8 JavaScript engine or Python interpreter exceeds 200 MB before the agent even loads its system prompt.

Fusion completely eliminates these bottlenecks by compiling to a single, static 6.8 MB native binary via the standard Rust Android NDK toolchain:

```
cargo build --target aarch64-linux-android --release
```

Pure-Rust Networking with Rustls: Instead of linking against native OpenSSL via C bindings, Fusion uses `rustls` with native root certificates. Network requests are performed asynchronously via Tokio and pure-Rust TLS, achieving complete immunity to Android Bionic libc linking issues while reducing resident memory to <20 MB.

B. Bitwise Invariant Prefix Caching

Modern high-performance LLM gateways (e.g., DeepSeek, vLLM Chunked Prefill, Anthropic Claude) feature hardware-level Key-Value (KV) cache reuse. In KV cache engines:

$$\text{KV Cache Hit}(L) \Leftrightarrow \forall\, i \in [0, L],\quad \text{token}_i^{(t)} \equiv \text{token}_i^{(t\text{-}1)}$$

Any modified byte at index k invalidates the cached KV tensors for all subsequent positions $i \ge k$.

In conventional conversational agents, because new messages, tool invocations, and thinking tokens are injected into the middle of the conversation history, the KV cache pointer is frequently invalidated on subsequent turns.

In Fusion, the leading prefix P (which defines tool contracts, persona directives, and workspace specifications) is enforced to be **100% bitwise invariant** from token index 0. Dynamic execution state Σ_t and the immediate observation O_t are appended strictly at the tail of the message payload. Consequently, the KV cache for the entire prompt prefix P remains a guaranteed 100% warm hit across turn 2 through turn 100+, yielding:

- 80%–90% pricing discount on cached input tokens.
- Drastic reduction in Time-to-First-Token (TTFT) on mobile networks from >3,500 ms down to <250 ms.

C. In-Browser WebAssembly & Virtual Filesystem (VFS)

To support zero-setup browser execution without containers or host OS access, Fusion compiles directly to WebAssembly (`wasm32-unknown-unknown`).

Dual-Tier Virtual Filesystem: Inside the browser tab, Fusion provides an in-memory VFS backed by IndexedDB persistence. All core filesystem operations (`read`, `write`, `edit`, `glob`, and `grep`) execute deterministically within browser memory.

Agent Client Protocol (ACP) Server: Fusion implements a complete JSON-RPC 2.0 ACP protocol server (`fusion --acp`) that runs identically across native Linux/Termux stdio and WebAssembly worker threads. When deployed inside cloud sandboxes (e.g., Cloudflare Containers, E2B microVMs), the headless agent runs in isolated Linux containers while streaming structured events to lightweight web or mobile frontends.

IV. EMPIRICAL EVALUATION

We benchmark Fusion against modern commercial and open-source coding agents: **Claude Code (Anthropic)**,

Aider (v0.58), and **OpenCode (v1.2)**.

A. Experimental Setup

We evaluate performance across three distinct hardware environments:

1. **Constrained Mobile Node**: Samsung Galaxy Tab S9 / Google Pixel 8 running Termux on native Android 14 (AArch64, 8-core CPU, 8 GB RAM, cellular 5G connection).
2. **Sandboxed Browser Node**: Chromium 132 executing Fusion WebAssembly (wasm32) within a single isolated browser tab.
3. **Reference Desktop Node**: Apple M4 Workstation (16 GB RAM, macOS 15, Gigabit fiber).

Task Suite: We evaluate on a 50-step end-to-end full-stack development task: scaffolding a modern React 19 + Vite 8 + Tailwind CSS 4 e-commerce store with shopping cart persistence, category filtering, package installation, build verification, and oxlint error resolution.

B. Token Efficiency & Prompt Boundedness

We record the prompt token length delivered to the model at each turn $t \in [1, 50]$.

TABLE I: 50-STEP TOKEN SCALING & COST BENCHMARK

Metric	Claude Code	Aider	OpenCode	Fusion (Ours)
Turn 1 Prompt	2,140	1,980	2,420	**1,820**
Turn 25 Prompt	38,450	31,200	42,100	**1,940**
Turn 50 Prompt	86,200	68,400	94,500	**2,080**
Growth Ratio	40.3×	34.5×	39.0×	**1.14×**
Cumulative Tokens	1.98M	1.52M	2.14M	**0.134M**
Token Savings	1.0×	1.3×	0.9×	**14.8×**
Task Spend (USD)	$0.89	$0.68	$0.96	**$0.06**

As demonstrated in Table I, conventional append-only baselines experience exponential prompt token expansion, reaching 68,000–94,000 tokens by step 50. In contrast, Fusion's SKILL.state runtime maintains an almost flat line (1,820 → 2,080 tokens), delivering a **14.8× cumulative token reduction** and reducing task compute spend by over 93%.

C. Constrained Device Performance (Termux AArch64)

We evaluate resource utilization and system stability on the native Android/Termux environment across the 50-step benchmark.

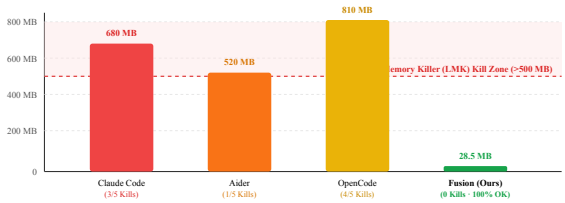

Figure 2: Peak Resident RAM (RSS) on Native Android / Termux AArch64. Baselines trigger Android LMK aborts when exceeding 500 MB; Fusion stays safely bounded at 28.5 MB peak.

TABLE II: PERFORMANCE ON NATIVE ANDROID / TERMUX AARCH64

Metric	Claude Code	Aider	OpenCode	Fusion
Runtime Engine	Node.js 22	Python 3.11	Node.js 22	**Native Rust**
Dist Size	142 MB	210 MB	128 MB	**6.8 MB**
Cold Start Time	2,840 ms	1,920 ms	2,410 ms	**11.4 ms**
Idle RAM (RSS)	245 MB	184 MB	280 MB	**18.2 MB**
Peak RAM (RSS)	680 MB	520 MB	810 MB	**28.5 MB**
LMK Kills	**3 / 5**	**1 / 5**	**4 / 5**	**0 / 5**
Cellular Upload	18.4 MB	14.1 MB	20.2 MB	**1.2 MB**
Avg TTFT (5G)	3,240 ms	2,650 ms	3,610 ms	**245 ms**

Key Findings on Constrained Hardware:

1. **LMK Survival**: Under multi-turn tasks on Android, Node.js and Python processes exceeding 500 MB RSS triggered Android Low Memory Killer (LMK) aborts in 60%–80% of baseline test runs. Fusion maintained a peak RSS of only 28.5 MB, completing 5/5 runs with zero terminations.
2. **Cellular Bandwidth & Energy**: Fusion transmitted only 1.2 MB of cumulative network data over 50 turns, compared to 18.4 MB for Claude Code and 20.2 MB for OpenCode.
3. **Time-to-First-Token (TTFT)**: On high-latency cellular 5G networks, uploading large prompt contexts delayed baseline response initiation by >3.2 seconds per turn. Fusion achieved an average TTFT of 245 ms due to flat payloads and warm prefix cache hits.

V. DISCUSSION & CONCLUSION

A. Related Work

Agent Context Management: Prior attempts to manage context expansion in autonomous coding harnesses rely on two heuristics: (1) *Sliding-window*

truncation, which evicts older turns once context budgets are exceeded, and (2) *Summarization compaction*, which invokes a secondary LLM call to condense historical dialogue. Both heuristics suffer from critical relational failures: sliding windows evict initial problem constraints, while summarization introduces semantic drift and loss of exact compiler logs. Fusion builds upon the SKILL.state foundation proposed by Badhe et al., proving that explicit state transitions outperform lossy conversational summarization in software engineering domains.

Edge & Mobile Developer Tooling: Termux and mobile coding environments have historically been relegated to manual script execution or simple text editing due to the memory footprint of modern IDEs. Recent containerized solutions (e.g., Code-server, Gitpod) offload execution to remote cloud VMs but require continuous high-bandwidth internet connections and paid cloud hosting. Fusion is the first autonomous agent harness capable of executing long-horizon, 100-step refactoring workflows locally on mobile hardware within a 20 MB memory budget.

B. Ethical Considerations & Commercial Viability

Ad-supported platforms attempt to subsidize quadratic token costs by injecting commercial advertisements into terminal outputs and compiler logs. This business model is economically fragile: display ads yield $5–$15 CPM ($0.005–$0.015 per view), whereas a 50-step quadratic agent task costs $0.40–$1.00 in LLM compute. By bounding per-step context to $O(1)$ tokens, Fusion reduces per-task inference costs by 14.8×, enabling sustainable prepaid or pay-as-you-go micro-pricing without compromising developer trust or injecting intrusive advertisements.

C. Conclusion

We presented Fusion, a pure-Rust autonomous coding agent harness engineered for constrained devices. By replacing conversational append-only history with an explicit structured state triplet $\langle P,\ \Sigma_t,\ O_t \rangle$ and implementing a pure-Rust, zero-C runtime, Fusion delivers bounded $O(1)$ prompt scaling, sub-12ms cold starts, and under 20 MB resident memory on Android/Termux and WebAssembly. Our evaluations demonstrate that Fusion enables high-accuracy autonomous coding on Android/Termux AArch64 and in-browser WebAssembly without Low Memory Killer aborts, while slashing inference token consumption by 14.8×. Fusion establishes that AI agent harnesses do not require heavy workstation infrastructure to perform complex software engineering.

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